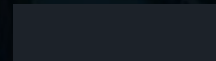
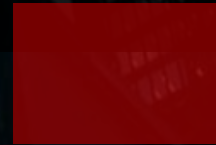




# Advanced Prototyping with Streamlit

---





# Streamlit in Snowflake

Architecture, capabilities, and where it fits in your AI stack



# AGENDA

Architecture & Capabilities

Why Streamlit for AI/BI?

UI Components & Layouts

State & Session Handling

4.5 Chat History & Context

Cortex & Data Connections

Persona-Aware Chatbot





# Streamlit for AI/BI?

The fastest path from prototype to production for AI-powered data apps

# Why Streamlit Wins for AI/BI



## Speed

Idea to demo in hours — no frontend framework, no build step



## AI-Native

Cortex LLM in 1 line — chat history in `st.session_state`



## BI-Ready

Native chart widgets + Altair/Plotly from any Snowflake query



## Secure

RBAC governs access — no secrets in code, no external auth



## Iterative

Live reload on save — stakeholder feedback on running app



## Deployable

One SQL command to production — same code, no re-architecture

# Tool Comparison

| Tool                   | Setup      | AI Integration  | Data Access    | RBAC     | Best For               |
|------------------------|------------|-----------------|----------------|----------|------------------------|
| Streamlit in Snowflake | Minutes    | Native Cortex   | Session object | ✓ Native | AI chatbots, BI agents |
| Sigma Computing        | Minutes    | Cortex formulas | Workbook/table | ✓ Native | Executive dashboards   |
| React + Node.js        | Hours/Days | REST API        | API layer      | Custom   | Production custom UI   |
| Jupyter Notebook       | Minutes    | Any library     | Connector      | None     | Data science / EDA     |
| Retool                 | Min–Hours  | REST API        | Connector      | Custom   | Internal admin tools   |

# Real-World Use Cases



## Event Performance Analyzer

NLQ chatbot — EA asks, Cortex translates to SQL



## Executive Revenue Briefing

Auto-generated bullet summaries — persona-aware C-suite format



## Attendee Sentiment Dashboard

CSV upload → CLASSIFY\_TEXT → Cortex SUMMARIZE findings



## Sponsor ROI Explorer

Interactive filters + AI explanation of metric changes



## Registration Anomaly Detector

Monitors live reg data, Cortex explains spikes/drops



## Persona-Aware BI Agent

Single app detects role → analyst view or executive view

✓ All 7 use cases run inside Snowflake — data never leaves the platform

# UI Components & Layout Patterns

Streamlit widgets, layout primitives, and AI-specific UI patterns





# Core Widgets for AI/BI Apps



## **st.chat\_message**

Chat bubble UI — role='user' or 'assistant', markdown-rendered



## **st.chat\_input**

Sticky input box at bottom — standard chat UX pattern



## **st.selectbox / radio**

Persona selector, model picker, time range controls



## **st.plotly\_chart**

Render Plotly/Altair/Vega charts from Snowflake results



## **st.dataframe**

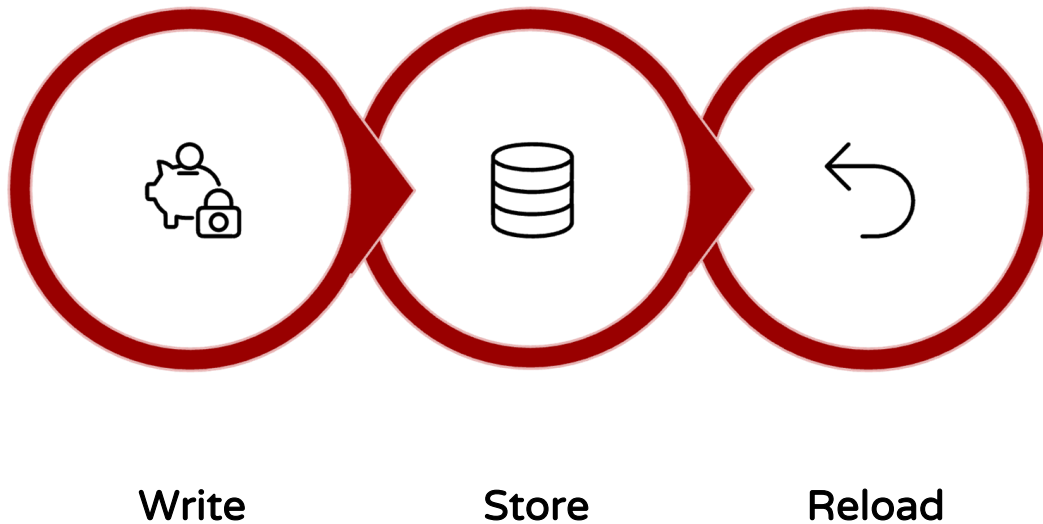
Interactive, sortable table — show SQL results with formatting



## **st.sidebar**

Navigation panel for filters, persona toggle, session controls

# Persisting Sessions to Snowflake



## Why persist to Snowflake?

- Conversation analytics
- Model comparison audits
- Compliance and governance
- Session restore across logins

✓ Full audit trail — who asked what, when, with which model

# Cortex Model Comparison

| Model                 | Context Window | Speed     | Best For                   | Cost |
|-----------------------|----------------|-----------|----------------------------|------|
| <b>mistral-large2</b> | 128K tokens    | Medium    | Complex NLQ, multi-turn    | \$\$ |
| llama3-70b-instruct   | 8K tokens      | Fast      | Short queries, summaries   | \$   |
| llama3-8b-instruct    | 8K tokens      | Very Fast | Formatting, classification | ¢    |
| snowflake-arctic      | 4K tokens      | Fastest   | Structured output          | ¢    |

## Model selection rule of thumb

- **Complex NLQ** → Mistral Large
- **Simple formatting** → Llama 3 8B
- **Structured output** → Arctic
- **Long documents** → mistral-large2

 Start with mistral-large2, optimize to smaller models after testing



# Building a Persona-Aware Chatbot

Dynamic routing, persona-specific prompts, and adaptive UX



A low-angle, upward-looking photograph of several modern skyscrapers with glass facades. The image is overlaid with a semi-transparent dark grey horizontal band across the middle. Three solid red rectangular blocks are positioned: one at the top center, one at the bottom left, and one at the bottom right. The text 'THANK YOU' is centered within the dark band in a white, bold, sans-serif font.

**THANK YOU**